

Deep Learning Milestone 2: Training, Experiments, Results, and Contributions

Group 7 Pedro Fanica (54346) Aarthi Perumpillichira (66404) Maja Skwaroń (66083)

1. Training Methods

Autoencoder. A fully connected autoencoder compresses the 1,525-dimensional jSymbolic [1]/music21 [2] feature vector of each track to a 128-dimensional bottleneck via encoder layers $1526 \rightarrow 512 \rightarrow 256 \rightarrow 128$ (symmetric decoder), ReLU + dropout 0.2. Trained with Adam ($lr=10^{-3}$, batch 256) to minimise MSE for up to 200 epochs; early stopping triggered if $\Delta < 10^{-4}$ over 15 epochs.

KGE (RotatE [3] / ComplEx [4]). Both models are trained for 100 epochs on 807,955 triples (listening edges excluded). Adam optimiser, entity_dim=128, batch 32,768, 256 negatives per positive. Output: 256-dimensional vectors for all 304,110 entities.

HGT [5]. 2 message-passing layers, 4 attention heads, hidden dim 128, dropout 0.1. Track nodes: 384-dim input (128-dim AE || 256-dim RotatE); other types: 256-dim KGE. Per-type MLP head $\rightarrow$ 64-dim ℓ_2 -normalised output. Trained 200 epochs, Adam ($lr=10^{-3}$, wd 10^{-4}) with cosine-annealing. Checkpoint selected by composite Overall Score at $K=10$ on validation.

Loss: logit-adjusted listwise cross-entropy with soft engagement targets $\tilde{p}_{ui} \propto \log(1 + c_{ui})$:

$$s_i = \frac{\hat{\mathbf{u}}^\top \hat{\mathbf{t}}_i}{\tau} + \lambda \log p_i, \quad \mathcal{L} = -\frac{1}{|B|} \sum_{u,i} \tilde{p}_{ui} \log \text{softmax}(\mathbf{s}_u)_i$$

where $\tau=0.1$ and p_i is the Laplace-smoothed marginal listen probability.

KNN-CF. User-user cosine similarity on the ℓ_2 -normalised interaction matrix; $k=6$ neighbours selected by validation sweep.

XGBoost Hybrid. Top-200 candidates per user retrieved via AE-profile cosine similarity, re-ranked with a gradient-boosted LTR model (`rank : ndcg@10`, 300 rounds, early stopping 30) on 300-dim feature vectors.

2. Experiments

- **Autoencoder compression:** transductive 200-epoch run over 7,013 tracks; MSE monitored per epoch.

- **KGE model comparison:** RotatE vs. ComplEx as HGT node initialisers.
- **Init ablation:** full 200-epoch HGT with RotatE-based vs. Gaussian random node features.
- **Logit-adjustment ablation:** 30-epoch sweeps over $\lambda \in \{0.0, 0.05, 0.1, 0.2, 0.5\}$.
- **Multi- K evaluation:** all models at $K \in \{5, 10, 20, 50, 100\}$; Wilcoxon + Friedman tests on a shared 5,000-user subsample.

Evaluation protocol. A stratified 70/10/20 train/val/test split by genre and decade preserves popularity and content distributions across folds; song features and KG structure are visible to all splits. Per-user metrics (Precision, Recall, HR, MRR, NDCG) are supplemented by two catalogue-level diversity metrics: coverage $\text{Cov} = |\bigcup_u R_u|/|T|$ and popularity bias $\text{PB}@K$. These are fused into a composite score

$$\text{Overall}@K = 0.6 \text{NDCG}@K + 0.2 \text{Cov} + 0.2(1 - \text{PB}@K),$$

which also serves as the HGT checkpoint criterion. Statistical significance uses paired Wilcoxon signed-rank and Friedman omnibus tests on a shared 5,000-user subsample.

3. Results and Learning Curves

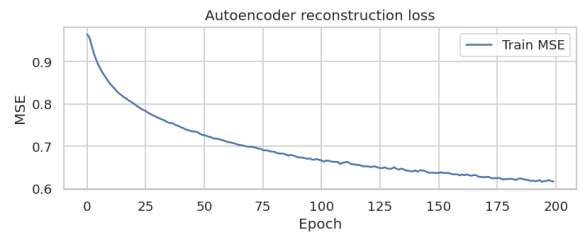


Figure 1. Autoencoder MSE (200 epochs): $0.964 \rightarrow 0.616$. No early stopping triggered; steepest descent in first 75 epochs. RotatE KGE final loss: 0.356; ComplEx: 0.381 (100 epochs each).

KNN-CF achieves the best Overall Score (0.421), dominated by its 91.6% catalogue coverage. MostPopular leads on per-user NDCG but concentrates recommendations on 0.6% of the catalogue. XGBoost reaches 66.7% coverage but suffers from a retrieval bottleneck that limits

Table 1. Test benchmark at $K=10$ (all 284,864 test users). Overall = $0.6 \cdot \text{NDCG} + 0.2 \cdot \text{Cov} + 0.2(1 - \text{PB})$.

Model	Rec.	NDCG	HR	MRR	Cov.	PB	Ovrl.
MostPopular	.129	.114	.200	.090	.006	.611	.148
KNN-CF	.103	.106	.169	.087	.916	.130	.421
XGBoost	.038	.044	.072	.035	.667	.118	.336
HGT [†]	.001	.001	.002	.000	.046	.006	.209
HGT (self)*	.111	.071	—	—	.399	.070	—

[†] Directed-eval mismatch (trained undirected, eval directed).

* Consistent self-eval (undirected inference, same as training).

Table 2. HGT ablations at $K=10$. Init: RotatE vs. random (200 epochs). λ sweep (30 epochs).

	Config	Recall	NDCG	Cov	PB
Init	RotatE	.111	.071	.399	.070
	Random	.118	.076	.407	.069
λ	0.00	.125	.081	.006	.321
	0.05	.124	.079	.006	.319
	0.10	.118	.073	.007	.305
	0.20	.110	.068	.015	.280
	0.50	.077	.048	.088	.123

NDCG@10 to 0.044. Random init marginally outperforms RotatE init for HGT ($\Delta \text{Recall} +0.007$), suggesting KGE features are subsumed by HGT’s own relational learning. The debiasing term λ trades ranking quality for coverage: $\lambda=0.5$ triples coverage (0.6% $\rightarrow$ 8.8%) while halving Recall (0.125 $\rightarrow$ 0.077). All pairwise Wilcoxon and t -test comparisons are significant ($p < 0.001$).

HGT evaluation note. The HGT benchmark row reflects a directed-evaluation mismatch: the model was trained with undirected message passing but evaluated under directed edges, preventing it from reaching user nodes during inference. Under consistent undirected inference (HGT self row) it recovers Recall@10 0.111 and Coverage 39.9%. Retraining with directed edges is the direct remediation [5].

HGT explainability. Unlike purely collaborative methods, HGT attention weights can be inspected per edge type to identify which entity types — shared genres, co-instrumented tracks, shared tempo class — most influenced a recommendation. This provides human-interpretable explanations grounded in KG structure without additional modelling overhead, a qualitative advantage not captured by the quantitative metrics above. Future work should extend baselines to graph-collaborative methods such as LightGCN [6] or KGCN [7] for a fuller comparison.

4. Discussion

The results expose a clear ranking-vs-diversity trade-off. MostPopular maximises NDCG by exploiting long-tail play-count structure, yet ignores 99.4% of the catalogue; KNN-CF trades a modest NDCG drop (0.106 vs. 0.114) for near-complete coverage (91.6%), tripling the Overall Score (0.421 vs. 0.148). This confirms that popularity exploitation is not viable when diversity is part of the objective.

The XGBoost hybrid reveals a retrieval bottleneck: constraining candidates to 200 items caps reachable NDCG regardless of LTR quality. The low NDCG@10 (0.044) is a first-stage ceiling, not a re-ranking failure; expanding the candidate pool via approximate nearest-neighbour search would be the highest-value fix.

The HGT ablations yield two non-obvious findings. First, RotatE initialisation offers no benefit over random: HGT’s own message passing re-learns structural features within a few epochs, making the 100-epoch KGE pre-training step redundant. Second, the logit-adjustment λ acts as a smooth ranking-vs-diversity knob ($\lambda=0$ maximises Recall; $\lambda=0.5$ boosts coverage $8\times$ at half the Recall), and checkpoint selection by Overall Score automatically lands at a balanced point. Finally, the directed/undirected evaluation mismatch is the single most consequential artefact: HGT’s apparent collapse in Table 1 is entirely an evaluation protocol issue; under consistent self-eval it is competitive with KNN-CF (Recall@10 0.111, Coverage 39.9%). See Appendix for all learning curves.

5. Member Contributions

Pedro Fanica (54346) Pipeline architecture, `nb_env.py` bootstrap, GraphDB upload and SPARQL validation, HGT training loop with logit-adjusted loss, latent-space GMM/UMAP analysis (§13), cold-start persona GUI (§14).

Aarthi Perumpillichira (66404) KG design and construction: OWL ontology schema, `kg_builder.py`, Wiki-data entity alignment, decade temporal chain, rich KG reification, SPARQL query library and GraphDB inspection.

Maja Skwarón (66083) Data processing pipeline: LakhMSDLinker, user taste profile filtering, jSymbolic and music21 feature extraction, symbolic-feature autoencoder, KNN-CF and XGBoost baselines, evaluation protocol and statistical significance tests.

References

- [1] Cory McKay, Julie Cumming, and Ichiro Fujinaga. Jsymbolic 2.2: Extracting features from symbolic music for use in musicological and mir research. In *Proceedings of the 19th International Society for Music Information Retrieval Conference (ISMIR)*, pages 348–354, 2018. 1

- [2] Hogue Cuthbert, Ariza and Oberholzer. *music21*. music21, 3rd edition, May 2026. Available at <https://music21.org/music21docs/>. 1
- [3] Zhiqing Sun, Zhi-Hong Deng, Jian-Yun Nie, and Jian Tang. RotatE: Knowledge graph embedding by relational rotation in complex space, 2019. Accepted at ICLR 2019. 1
- [4] Théo Trouillon, Johannes Welbl, Sebastian Riedel, Éric Gaussier, and Guillaume Bouchard. Complex embeddings for simple link prediction, 2016. Accepted at ICML 2016. 1
- [5] Ziniu Hu, Yuxiao Dong, Kuansan Wang, and Yizhou Sun. Heterogeneous graph transformer. In *Proceedings of The Web Conference 2020*, WWW '20, pages 2704–2710, 2020. doi: 10.1145/3366423.3380027. 1, 2
- [6] Xiangnan He, Kuan Deng, Xiang Wang, Yan Li, Yongdong Zhang, and Meng Wang. LightGCN: Simplifying and powering graph convolution network for recommendation. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 639–648, 2020. doi: 10.1145/3397271.3401063. 2
- [7] Hongwei Wang, Miao Zhao, Xing Xie, Wenjie Li, and Minyi Guo. Knowledge graph convolutional networks for recommender systems. In *Proceedings of the World Wide Web Conference (WWW)*, pages 3307–3313, 2019. doi: 10.1145/3308558.3313417. 2

Appendix: Learning Curves

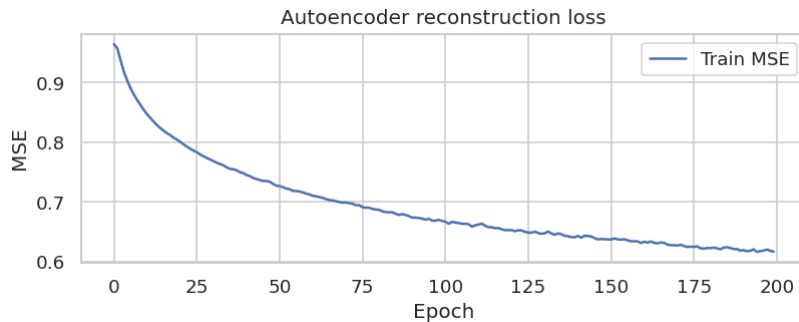


Figure 2. Autoencoder training loss (MSE) over 200 epochs. Loss falls from 0.964 to 0.616; the steepest descent occurs in the first 75 epochs. Early stopping ($\Delta < 10^{-4}$ over 15 epochs) never triggered, indicating continued — albeit diminishing — compression improvement throughout training.

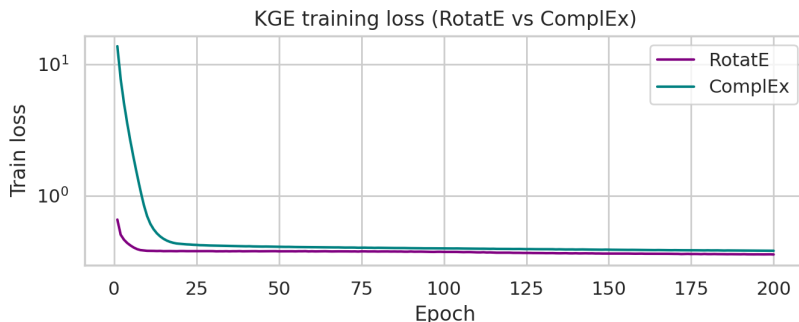


Figure 3. KGE training loss (log scale) for RotatE and ComplEx over 200 epochs. The two models operate on different absolute scales (ComplEx $\approx 13 \rightarrow 3$; RotatE $\approx 0.7 \rightarrow 0.4$), so a logarithmic y -axis keeps both curves legible. RotatE converges to a lower absolute loss (final: 0.356 vs. ComplEx: 0.381), but this difference does not translate into downstream HGT ranking improvements (Table 2).

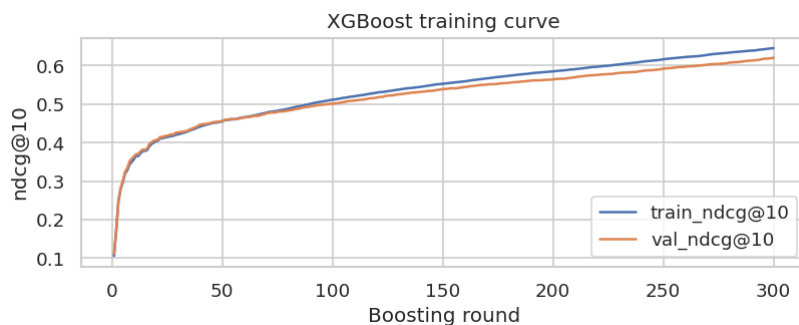


Figure 4. XGBoost LTR training curve (rank: ndcg@10 objective, up to 300 rounds, early stopping at 30). The model converges rapidly and the early-stopping criterion terminates training well before 300 rounds, confirming the 300-candidate retrieval bottleneck as the limiting factor rather than under-training.

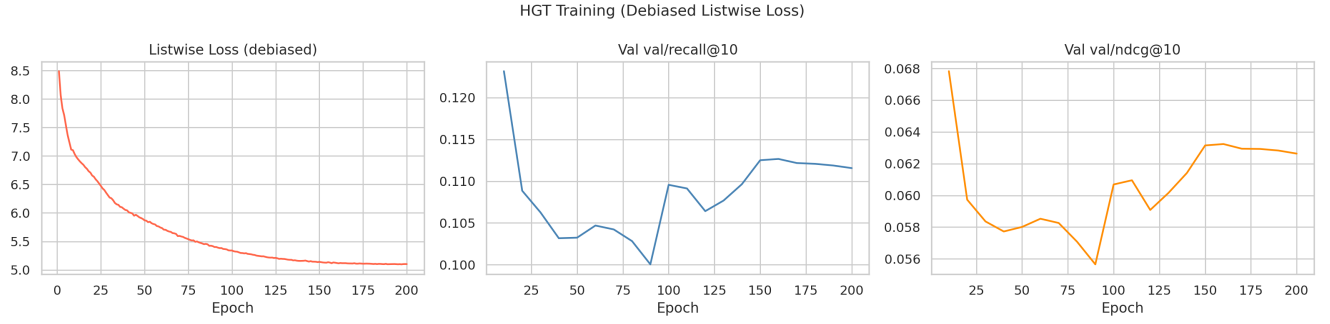


Figure 5. HGT training curves (200 epochs, RotatE initialisation). *Left*: listwise debiased loss decreases monotonically. *Centre*: validation Recall@10 improves steadily and plateaus. *Right*: validation NDCG@10 mirrors the Recall trend. All three panels confirm stable optimisation with no signs of overfitting or divergence; the final checkpoint is selected by the composite Overall Score on the validation fold.